ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A5 · FOOD CHAINS

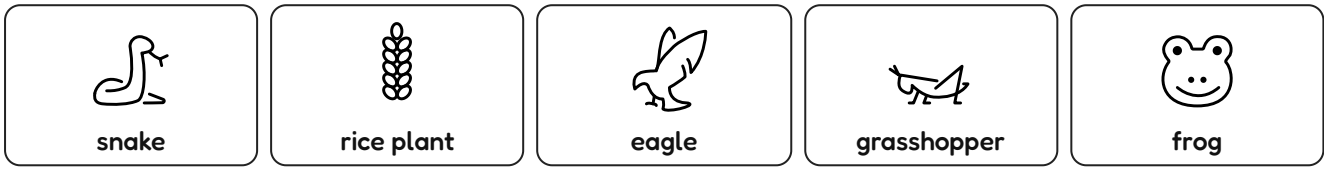
Who eats whom?

★☆☆
Grade 4

Learning goals

- Build food chains.
- Use food chain vocabulary.

1 Put the living things in order to make a food chain. Draw the arrows. [3]



rice plant → grasshopper → frog → snake → eagle

2 Use the food chain in question 1. [4]

Question	Answer
Which is the producer?	rice plant
Which is a herbivore?	grasshopper
Name a predator and its prey.	frog (predator) and grasshopper (prey)
Where does the energy come from first?	the Sun

3 Sort the animals. [9]

SORT THESE

caterpillar tiger deer cow person monkey crow crocodile kingfisher

Herbivore	Carnivore	Omnivore
cow	tiger	crow
deer	kingfisher	monkey
caterpillar	crocodile	person

4 **True or false?** [3]

Statement	True	False	If false, correct it
1. Every food chain starts with a plant.	✔		
2. The arrow in a food chain points to the food.		✔	The arrow points from the food to the animal that eats it.
3. An animal can be both a predator and prey.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A5 · FOOD CHAINS

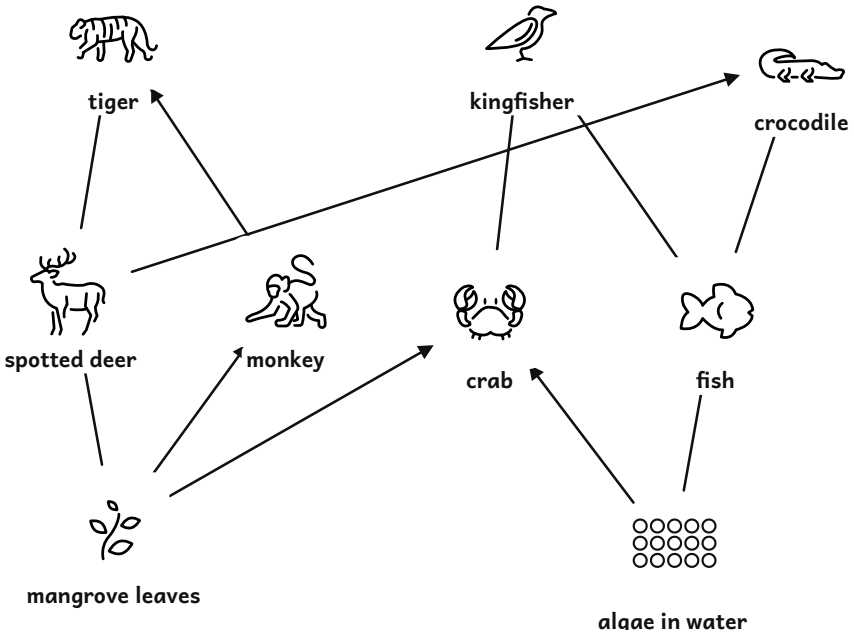
Exploring a food web

★★☆
Grade 4

Learning goals

- Read a food web.
- Predict effects of removing a living thing.

1 **Look at the Sundarbans food web.** [7]



a) Write two food chains from the web.

mangrove leaves → spotted deer → tiger; algae → fish → crocodile (or algae → crab → kingfisher)

b) Name an animal that eats more than one kind of food.

tiger (deer and monkeys) or kingfisher (crabs and fish) or crab

c) What would happen to the kingfishers if all the fish died? Explain.

They would have less food; they would eat more crabs, so crab numbers might fall, or some kingfishers might leave or die.

d) What would happen to the crabs if all the kingfishers died?

There might be more crabs because fewer are being eaten.

2 **Why do almost all food chains begin with a green plant?** [2]

Only plants can make their own food using energy from sunlight; all the energy in the chain starts there.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · A5 · FOOD CHAINS

Counting a rice field food chain

★★☆
Grade 4

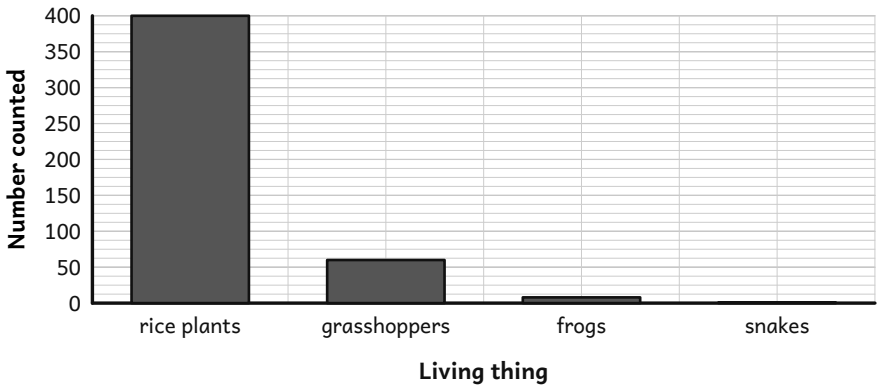
Learning goals

- Record and present survey data.
- Spot a pattern.
- Explain the pattern.

Scientists counted the living things in one small part of a rice field (example data).

1 **Draw a bar chart.** [3]

Living thing	rice plants	grasshoppers	frogs	snakes
Number counted	400	60	8	1



2 **Describe the pattern as you go along the food chain.** [2]

There are fewer and fewer living things at each step of the food chain.

3 **Suggest why there are so few snakes.** [2]

Each snake needs to eat many frogs to get enough energy, so the field can only feed a few snakes.

4 **Farmers sprayed a poison that killed most of the frogs. Predict what would happen to:** [3]

a) the grasshoppers

They would increase because fewer are eaten.

b) the rice plants

They would be damaged because more grasshoppers eat them.

c) the snakes

They would go hungry and their numbers would fall.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A5 · FOOD CHAINS

When the vultures disappeared

★★★
Grade 4

Learning goals

- Use a real food web problem.
- Interpret data about a falling population.
- Evaluate solutions.

Science | Health | Maths | Citizenship

Vultures are large birds that eat the bodies of dead animals. They are nature's cleaners: a group of vultures can clean a dead cow in less than an hour, stopping germs from spreading.

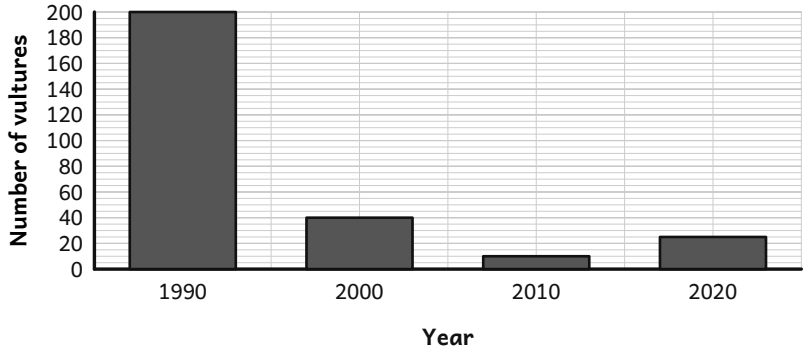
Since the 1990s, the number of vultures in South Asia has fallen by **more than 95 out of every 100**. Scientists found the cause: a medicine called **diclofenac** that was given to sick cattle. When a treated cow died and vultures ate it, the medicine poisoned the vultures.

With fewer vultures, dead animals lay rotting for longer. More stray dogs and rats came to eat them, and diseases can spread more easily. Bangladesh **banned diclofenac** for animals in 2010 and set up **Vulture Safe Zones** in 2014, where vultures are fed safe food and the medicines are checked.

Based on research by vulture conservation groups (e.g. IUCN Bangladesh, SAVE). Numbers below are example data.

1 **Vultures counted at one site (example data).** [3]

Year	1990	2000	2010	2020
Vultures	200	40	10	25



a) Draw the bar chart.

b) What happened after 2010? Suggest why.

The number went up a little, because diclofenac was banned and safe zones were made.

2 **Complete the chain of effects.** [3]

5	more stray dogs and rats, and more disease
1	farmers give sick cattle diclofenac
4	dead animals rot for longer
2	vultures eat dead cattle and are poisoned
3	fewer vultures

3 **Who can help vultures? Suggest one action for each.** [3]

Person	Action
a vet	use safe medicines such as meloxicam, not diclofenac
a farmer	leave dead cattle in safe places for vultures; use safe medicines
a learner	tell others why vultures matter; make a poster

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · A5 · FOOD CHAINS

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

V	N	U	W	S	E	P	L	S	U
H	C	C	H	A	I	N	E	R	O
M	O	P	L	A	N	T	S	W	C
U	N	S	E	N	E	R	G	Y	R
P	S	D	V	S	F	W	K	L	R
H	U	V	E	Y	W	G	T	W	S
W	M	P	R	E	D	A	T	O	R
E	E	P	R	O	D	U	C	E	R
B	R	S	U	N	P	R	E	Y	E
D	O	H	V	P	Y	H	C	T	T

chain consumer energy plant predator prey producer sun web

2 In the food chain grass → goat → tiger, which is the prey? [1]

- ○ grass
- ● **goat**
- ○ tiger
- ○ the Sun

3 What do all food chains start with? [1]

- ○ a predator
- ○ a herbivore
- ● **a producer**
- ○ a carnivore

4 Draw a food chain with four living things from a pond. Label the producer, a predator and its prey. [3]

My pond food chain

e.g. algae → small fish → big fish → kingfisher; algae = producer; kingfisher = predator, big fish = prey.

5 **Challenge. Explain why protecting the producers is the most important way to protect a whole food web.** [2]

All the energy in a food web starts with the producers. Without them the herbivores starve, then the carnivores, so the whole web collapses.